The Maintenance Item of Fiber Laser Cutting Machine

I:The daily checking item

II: The weekly or monthly checking item

III: The every three months maintenance item

IV:The every six months maintenance item

The daily checking item

1. Gas
2. Nozzle
3. Laser Head
4. Red light
5. Remote control handle
6. Anti-dust cover
7. The center position of the laser beam
8. Emergency devices
9. The small cart for collecting materials
10. The temperature of water chiller

The Daily Maintenance item

Item 1: gas



Name: the nitrogen/oxygen pressure gauge

Cycle: everyday

Method: check the pressure gauges to confirm if the values can meet the requirements.

1.8-2.3MPa for nitrogen pressure

0.15-0.6 MPa for oxygen pressure

The Daily Maintenance item

Item 2: nozzle



Name: nozzle

Cycle: everyday

Method: Counterclockwise rotate and take out the nozzle, look at the surface of nozzle to check if there is any damage, especially check if there is any slag at the surface of the nozzle and make sure the roundness is qualified.

The Daily Maintenance item

Item 3: Cable near laser head



Name: Cable near laser head

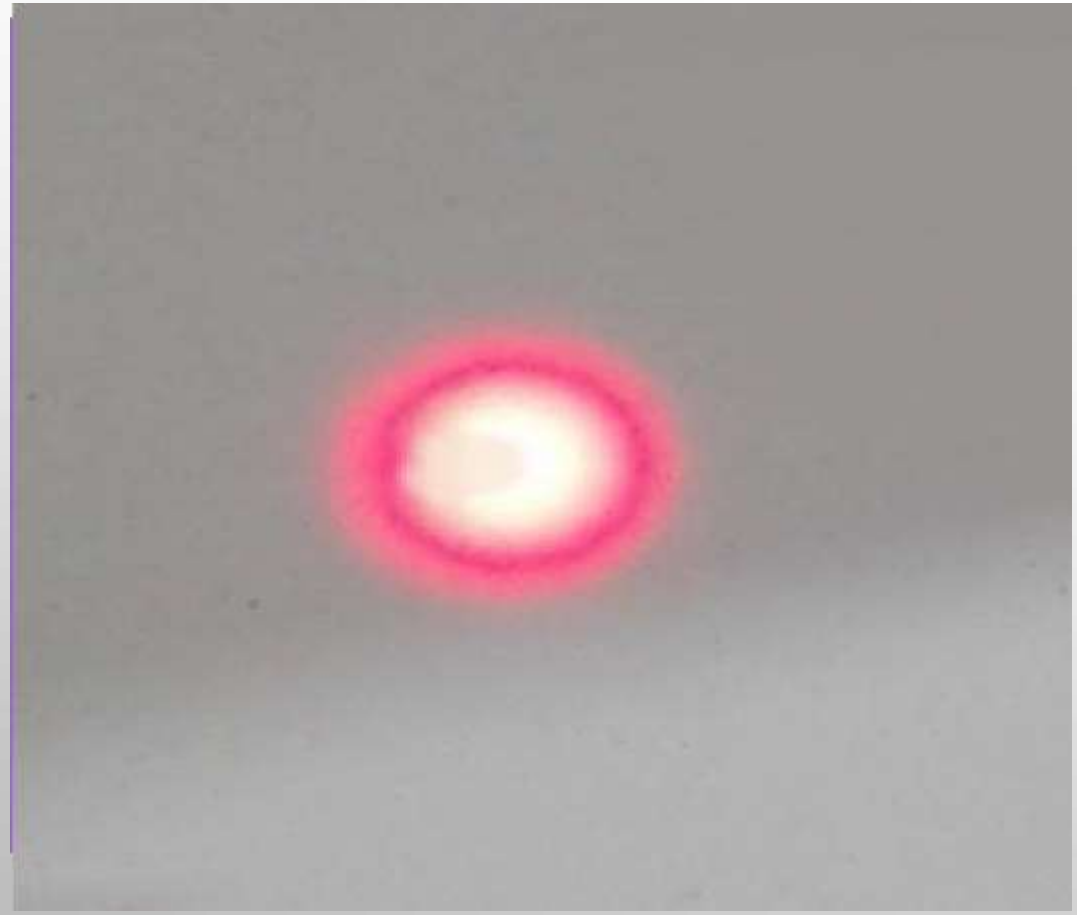
Cycle: everyday

Method: 1.Check the cable near the laser head, and banding the cable firmly with cable tie.

2.Inspection of circulating water lines, ensure no water leakage

The Daily Maintenance item

Item 4: red light



Name: the red light

Cycle: everyday

Method: Place one piece of white A4 paper under the laser head and obeserve with eyes whether there is black spot in the red light or not. If no black spots, the red light is qualified.

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The Daily Maintenance item

Item 5: remote control handle





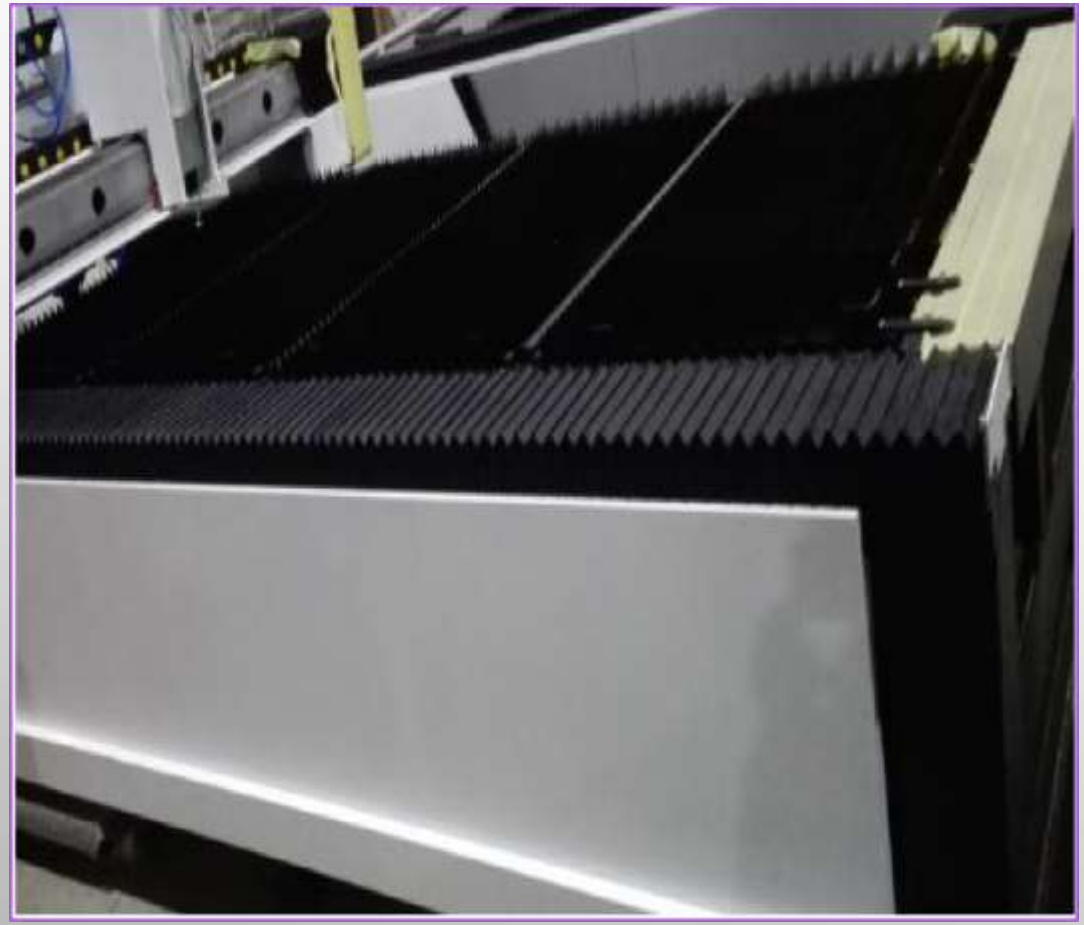
Name: the remote control handle

Cycle: everyday

Method: After start-up, press the button of control handle to check whether the buttons are sensitive or not. Also check whether the appearance is without any damage and the alarm light is in normal status.

The Daily Maintenance item

Item 6: anti-dust cover



Name: the anti-dust cover

Cycle: everyday

Method: Observe the surface of the anti-dust cover to make sure there is no damage and no extra dust. The extra substance is prohibited to enter the cover otherwise the precision of the guide rail and the gear will be affected.

The Daily Maintenance item

Item 7: the center position of the laser beam



Name: the laser beam

Cycle: everyday

Method: first paste the scotch tape under nozzle, then press the pulse button to check whether the laser beam is in the center or not. If the laser beam is not in the center, please readjust the laser beam.

The Daily Maintenance item

Item 8: the red emergency button



Name: red emergency button

Cycle: everyday

Method: Press the red emergency button of the displayer manually everyday to observe whether there is alarm, if not, then the red emergency button does not work normally.

The Daily Maintenance

Item 9: the small cart for collecting materials



Name: the small cart for collecting materials

Cycle: everyday

Method: Before off-duty everyday, please make sure there is no dust or abandoned materials in the small cart.

The Daily Maintenance item

Item10: the temperature of water chiller



Name: the temperature of water chiller

Cycle: everyday

Method: Turn on machine and run for a period of time, then observe the water chiller's display panel, making sure the high temperature lower than 30 degrees Celsius and the low temperature lower than 28 degrees Celsius .

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The weekly or monthly checking catalogue

1. The lubricating pump for the guide rail
2. The protective lens
3. The medium of water chiller
4. The water chiller
5. The rack
6. The guide rail
7. The gear and rack

The Weekly Maintenance item

Item 1: the lubricating pump for the guide rail



Name: the lubricating pump for the guide rail

The oil No.: No.68 lithium-based grease

Cycle: one week

Method: Check the lubricant oil level weekly to ensure the lubricant oil level is above MIN position. Add the lubricant oil in time if the lubricant oil is not enough .

The working pressure: 3bars .

The Weekly Maintenance item

Item 2: protective lens



Name: the protective lens

Cycle: each week

Method: Counterclockwise rotate the two bolts to take out the protective lens, then observe whether there is black or white spot onto lens.

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The Weekly Maintenance item

Item 3: the medium of water chiller



Name: the medium of water chiller

Cycle: each week

Method: Use the pure water that doesn't contain any minerals or use the steamed water or the deionized water (mineral water is prohibited).

The cooling water should be changed every 3-6 months.

The Weekly Maintenance item

Item 4: the filter gauge of water chiller



Name: the filter gauze of water chiller

Cycle: each week

Method: To ensure the cooling effect of the water chiller, the filter gauze need to be cleaned every week.

The every three months maintenance item

Item 1: the rack



Name: the rack

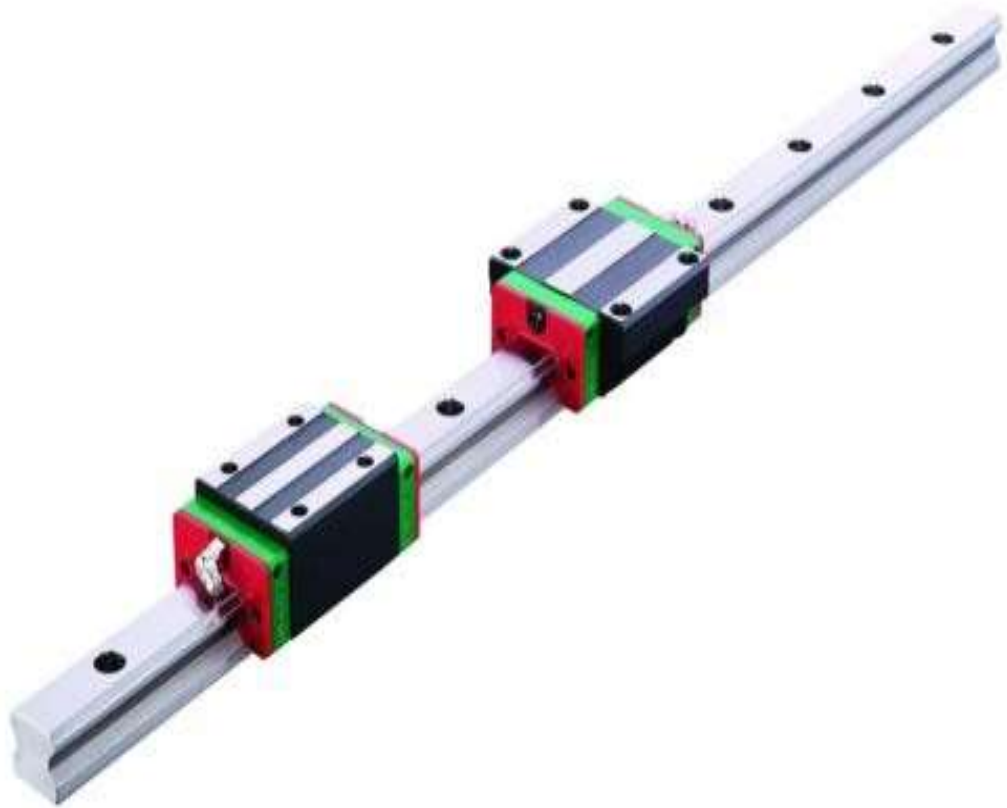
Cycle: every three months

Method: Observe the rack with eyes to check if there is any damage or rust. If the rack is rusty, anti-rust oil need to be smeared on it.

The type of anti-rust oil: WD-40

The every three months maintenance item

Item 2: the guide rail



Name: the guide rail

Cycle: every three months

Method: Observe the surface of the guide rail with eyes to check if there is any rust or oil stains. Use the clean brush to wipe the stains and then add the lubricant oil for the guide rail.

The maintenance every six months item

Item 2: Z shaft guide rail, guide rail slider, ball screw



Name: Z shaft guide rail, guide rail slider, ball screw

Cycle: every months

Method: 1.Check and clean Z shaft guide rail, guide rail slider, ball screw, Z shaft guide rail slider add lubricating oil manually, ball screw oil injection. (The guide rail slider oil storage box adds 32 machine tool guide oil, the ball screw selects 46 machine tool guide oil)

2.Clean and clean the inside and outside of the automatic lubricating container and replace the new lubricating oil.

3. Some Z-axis sliders have an automatic lubricating oil box, which can be filled with lubricating oil; some do not have this function, you need to apply grease on the guide rail

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The maintenance every six months item

Item 3: the meshing between gear and rack



Name: the gear

Cycle: every 6 months

Method: Check if there is any visible abrasion between the gear and rack, if the abrasion is very severe, the precision will be affected and the accessories should be changed.